Unsupervised Learning

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Unsupervised Learning is a type of Machine Learning where the model is trained using unlabeled data.

There is no target/output column.
The model tries to find hidden patterns, structures, or relationships in the data.

Why Do We Use Unsupervised Learning

We use unsupervised learning when:

- Data does not have labeled outputs
- We want to discover patterns
- We want to group similar data
- We want to reduce dimensionality
- We want to detect anomalies

Types of Unsupervised Learning

There are mainly two major types**:**

- Clustering
- Dimensionality Reduction

Clustering

Clustering is the process of grouping similar data points together.
Data points in the same cluster are similar.
Data points in different clusters are different.

Popular Clustering Algorithms

K-Means Clustering

- Divides data into K number of clusters
- Based on distance (usually Euclidean distance)
- Each cluster has a centroid

Steps

1. Choose K (number of clusters)

2. Select random centroids
3. Assign points to nearest centroid
4. Update centroid
5. Repeat until convergence

Example:

- Customer segmentation
- Grouping students based on marks

Hierarchical Clustering

- Builds a tree-like structure (dendrogram)
- Two types:
    - Agglomerative (bottom → up)
    - Divisive (top → down)

DBSCAN

- Density-based clustering
- Forms clusters based on density
- Good for irregular shapes
- Detects outliers

Dimensionality Reduction

It reduces the number of input features while keeping important information.

Used when dataset has many features

Reduces complexity
Improves visualization

Popular Dimensionality Reduction Techniques

PCA (Principal Component Analysis)

- Most popular technique
- Converts features into principal components
- Maximizes variance
- Linear method

Uses: t-SNE (t-Distributed Stochastic Neighbor Embedding)

- Non-linear method
- Best for visualization
- Preserves local structure

LDA (Linear Discriminant Analysis)

Technically supervised, but used for dimensionality reduction.

- Data visualization (2D/3D)
- Noise reduction
- Speed up training

Real-Life Applications

1. Customer Segmentation

E-commerce companies group customers based on buying behavior.

2. Market Basket Analysis

Finds products frequently bought together.

3. Image Compression

Reduce image size using PCA.

4. Anomaly Detection

Detect fraud transactions.

5. Recommendation Systems

Group similar users or products.

Advantages of Unsupervised Learning

- No labeled data required
- Finds hidden patterns
- Useful for exploration
- Works with large datasets

Disadvantages

- Hard to evaluate accuracy
- Results may be difficult to interpret
- Sensitive to noise